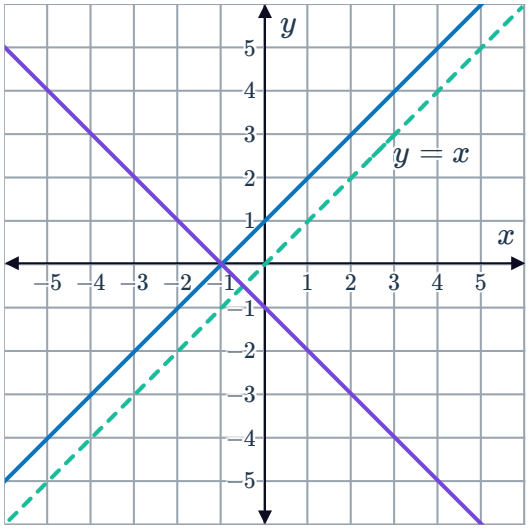
Inverse functions

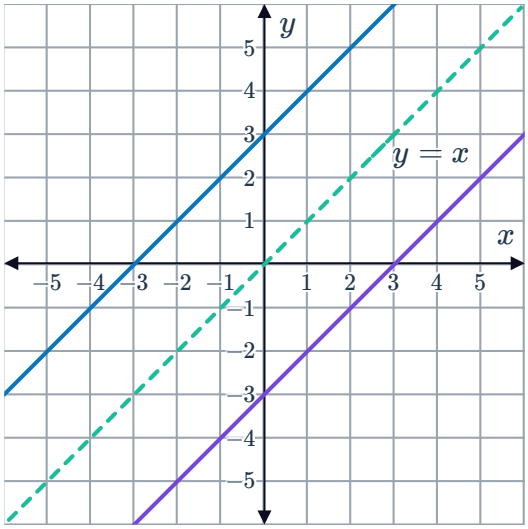
1 Explain how a graphing utility such as a graphing calculator be used to visually determine if two functions are inverses of each other.

2 In each graph below two functions and the line $y = x$ are drawn. For each graph, state whether the functions are inverse functions of each other.

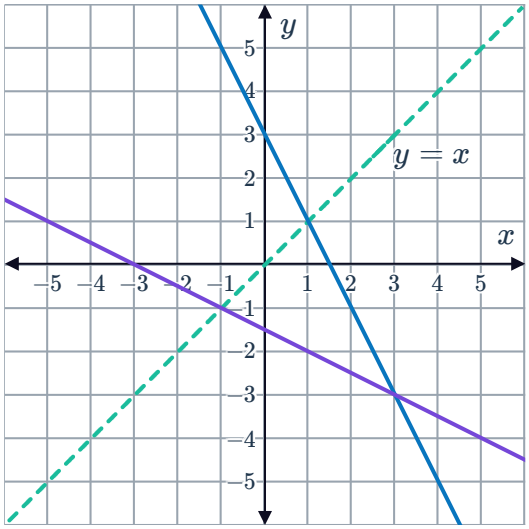
a



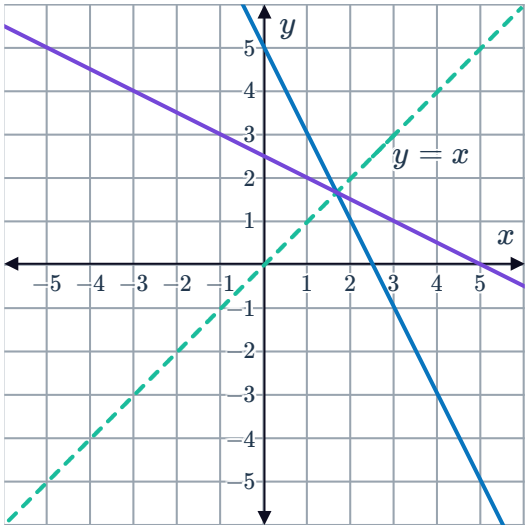
b



c

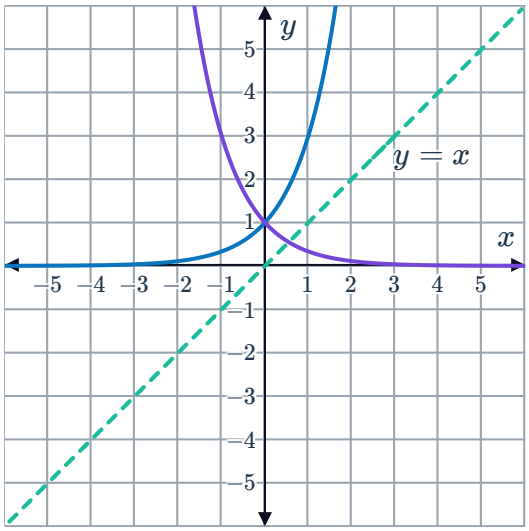


d

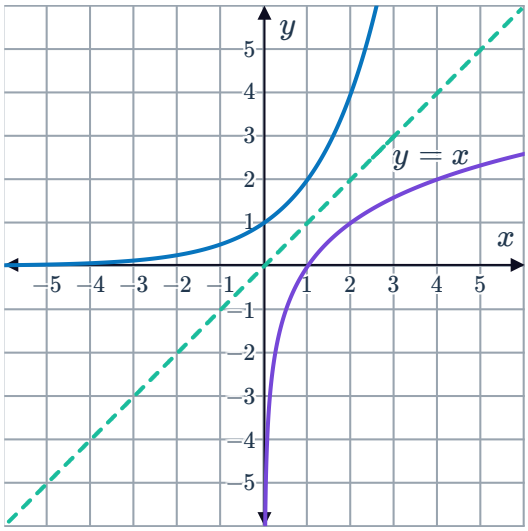


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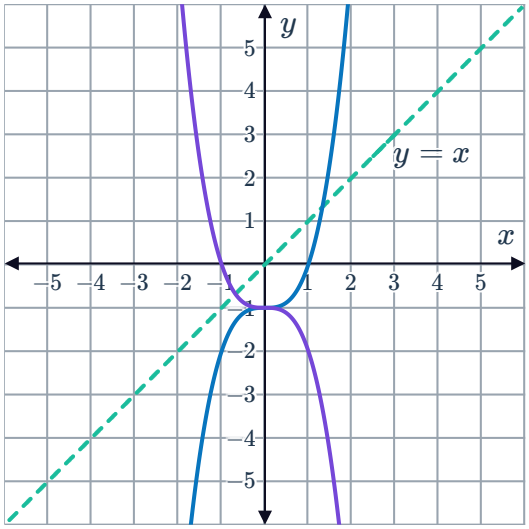
e



f



g



3 For each of the following functions, find:

i $f^{-1}(x)$

ii The domain of $f^{-1}(x)$

iii The range of $f^{-1}(x)$

a $f(x) = 8x - 9$ defined over $[-4, 2]$

b $f(x) = x^2$ defined over $[0, \infty)$

c $f(x) = \sqrt{16 - x^2}$ defined over $[0, 4]$

4 State whether the following functions have an inverse function:

a $f(x) = \dfrac{7}{x}$

b $f(x) = \sqrt{7 - x^2}$

c $f(x) = \sqrt{x} + 7$

d $f(x) = 1 - 7x$

e $f(x) = 7^x$

f $f(x) = 4x^3 + 8$

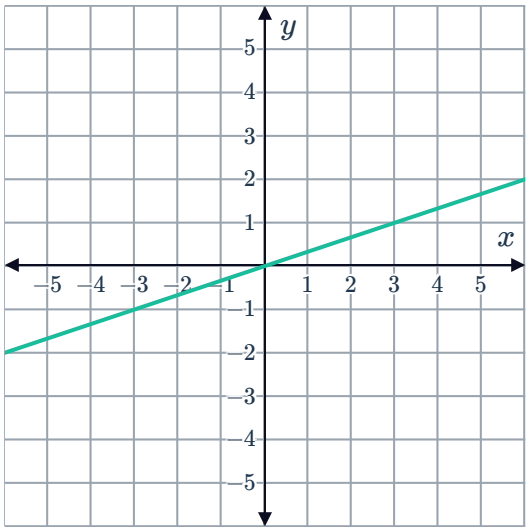
g $f(x) = \sqrt{4 + x}$

h $f(x) = -\dfrac{1}{7}x^2$

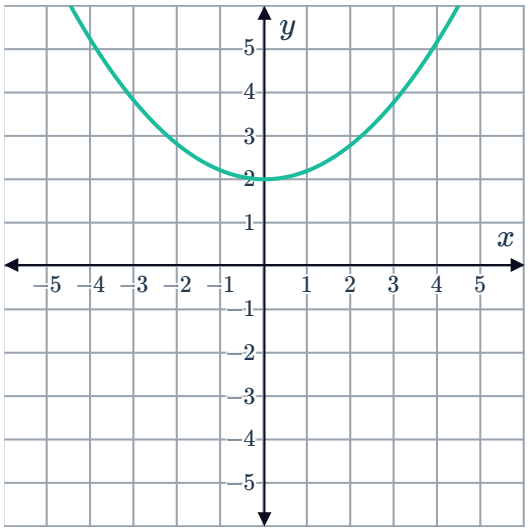
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5 State whether the following functions have an inverse function:

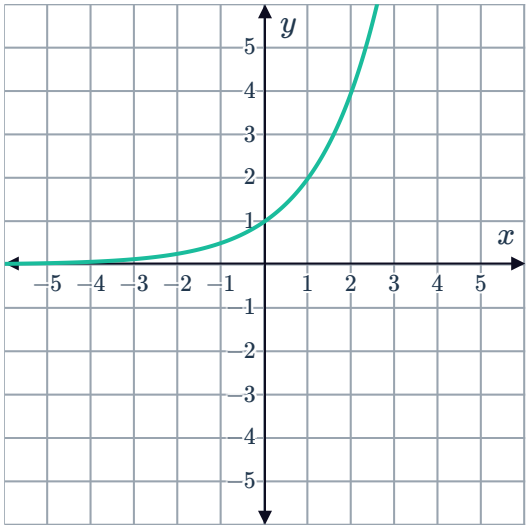
a



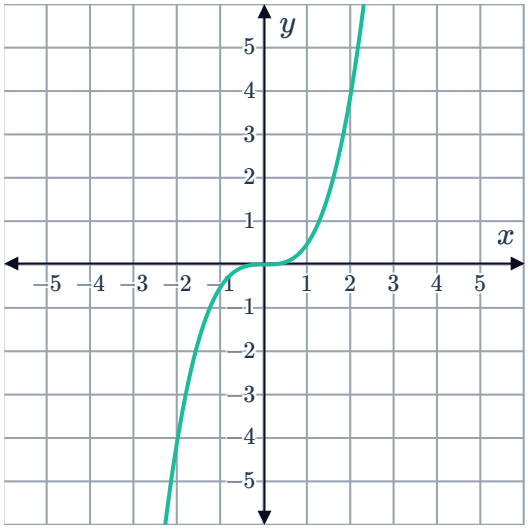
b



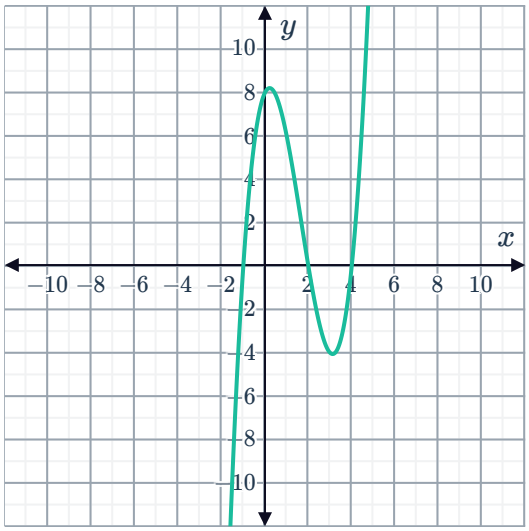
c



d



e



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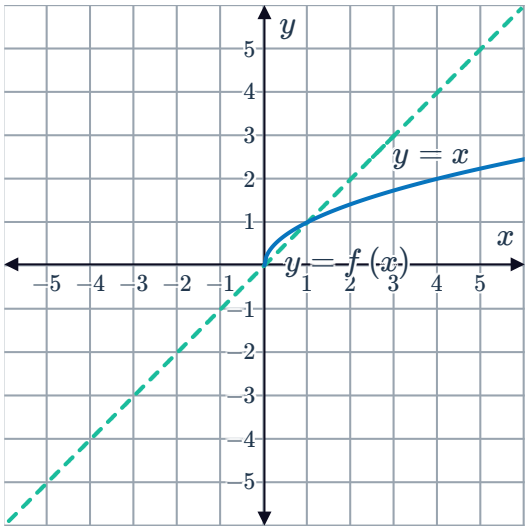
Graphs of inverse functions

6 Consider the functions $f(x) = \frac{1}{x} + 3$ and $g(x) = \frac{1}{x-3}$.

- **a** Sketch the graph of $f(x)$.
- **b** Sketch the graph of $g(x)$ on the same set of axes.
- **c** Are $f(x)$ and $g(x)$ inverses?

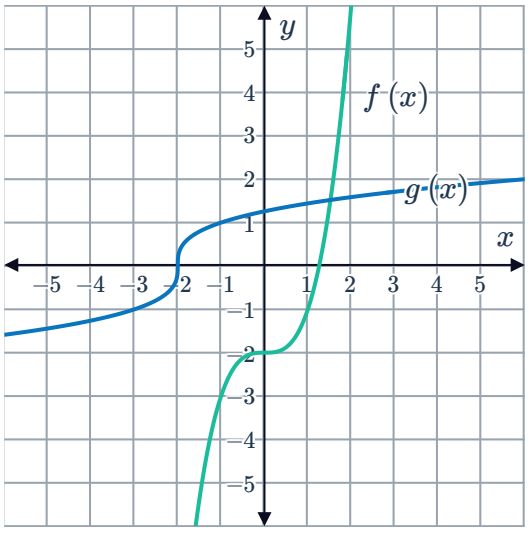
7 Consider the graph of the function $f(x)$ over the line $y = x$:

Sketch the graph of $f^{-1}(x)$.



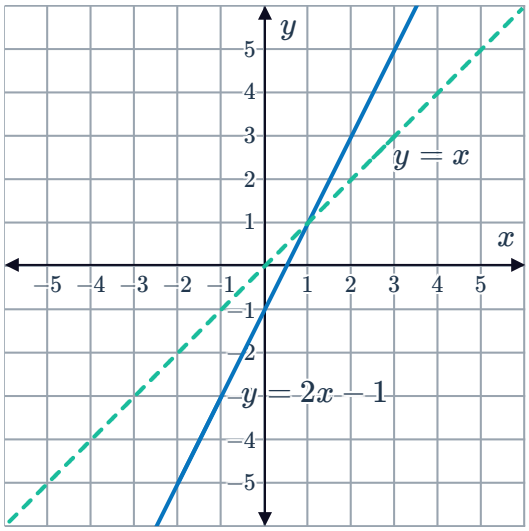
8 Consider the graphs of $f(x)$ and $g(x)$:

- **a** State the equation of $f(x)$.
- **b** State the equation of $g(x)$.
- **c** Evaluate $f(g(x))$.
- **d** Evaluate $g(f(x))$.
- **e** State whether the following statements are correct:
  - **i** $g(x)$ is an inverse of $f(x)$.
  - **ii** $f(g(x))$ has gradient -2.
  - **iii** $f(x)$ is an inverse of $g(x)$.
  - **iv** $g(f(x))$ has gradient 1.

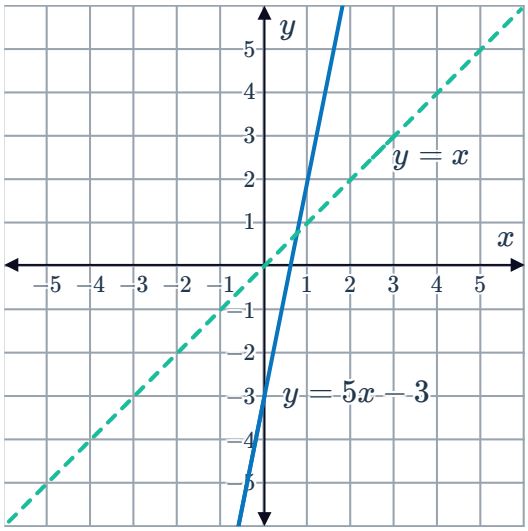


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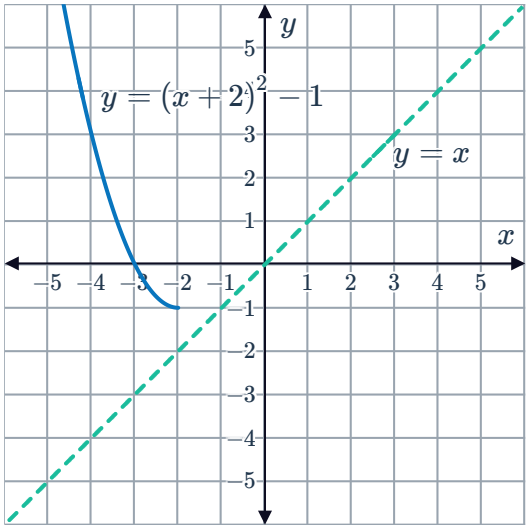
e



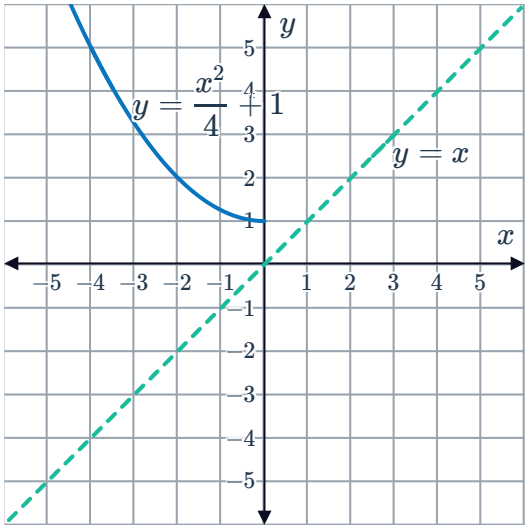
f



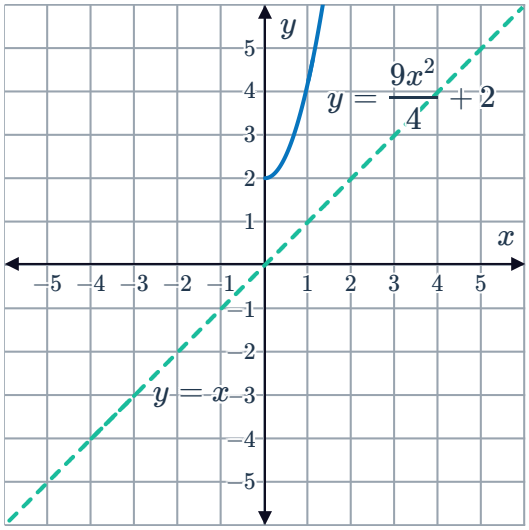
g



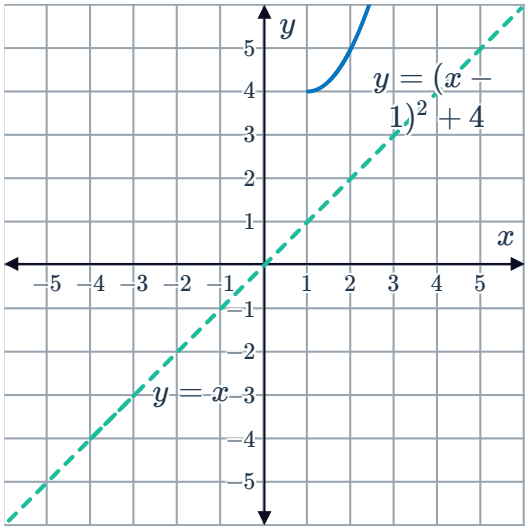
h



i



j



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11 For each of the following functions:

i Sketch the function $f(x)$ over its domain.

ii Find the inverse, f^{-1}.

iii State the domain of f^{-1}.

iii State the range of f^{-1}.

iv Sketch the function f^{-1} over its domain.

a $f(x) = x + 6$ defined over the interval $[0, \infty)$.

b $f(x) = 7 - x$ defined over the interval $[2, 9]$.

c $f(x) = (x - 6)^2 - 2$ defined over the interval $[6, \infty)$.

d $f(x) = \sqrt{4 - x}$ defined over the interval $[0, 4)$.

e $f(x) = (x + 2)^2 + 3$ defined over the interval $[0, \infty)$.

12 Consider the functions $f(x) = x^2 - 5$ and $g(x) = \sqrt{x + 5}$, for $x \geq 0$. The function y is defined as $y = g(f(x))$, for $x \geq 0$.

a State the equation for y.

b Graph the functions $f(x)$, $g(x)$ and y on the same set of axes.

c What do you notice about the graph of y in relation to the graphs of $f(x)$ and $g(x)$?

Applications

13 The function $t = \sqrt{\dfrac{d}{4.9}}$ can be used to find the number of seconds it takes for an object in Earth's atmosphere to fall d metres.

a State the function for d in terms of t.

b Find the distance a skydiver has fallen 5 seconds after jumping out of a plane.

14 The following formula can be used to convert Fahrenheit temperatures x to Celsius temperatures $T(x)$:

$$T(x) = \frac{5}{9}(x - 32)$$

a Find $T(-13)$.

b Find $T(86)$.

c Find $T^{-1}(x)$.

d What can the formula T^{-1} be used for?

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15 The function $d(t) = 120 - 4.9t^2$ can be used to find the distance, d, that an object dropped from a height of 120 metres has fallen after t seconds.

a Is the function $d(t)$ one-to-one?

b Find the inverse function, $t(d)$.

c Prove that $d(t(d)) = d$.

d Prove that $t(d(t)) = t$.

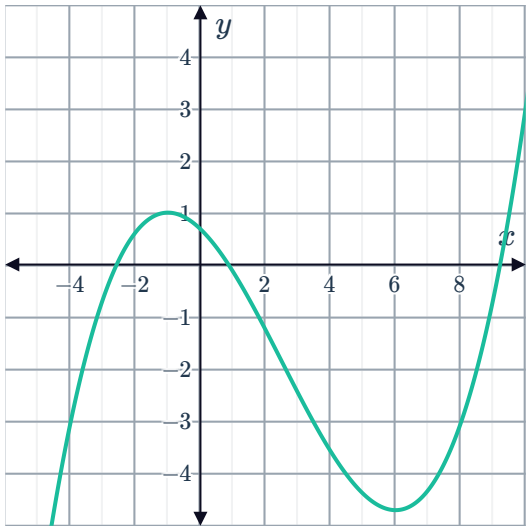
e Are d and t inverse functions?

f How long it will take an object to fall 41.6 metres when dropped from a height of 120 metres?

16

a Consider the function $f(x) = (x-2)^2 - 6$. Write the function as two one-to-one functions that have the same rule as $f(x)$, but specific domains.

b Consider the graph of $y = g(x)$ shown:



Complete the piecewise function $h(x)$ below to split the graph of $y = g(x)$ into three one-to-one functions that have the same rule as $g(x)$:

$$h(x) = \begin{cases} g(x), & x \leq \square \\ g(x), & \square < x < \square \\ g(x), & x \geq \square \end{cases}$$

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